

ANDREA D'AMBROSIO

Postdoctoral Research Associate, The University of Arizona

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🌐 LinkedIn 🌐 Systems & Industrial Engineering 🌐 Space4 Center
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EDUCATION

Ph.D. in Astronomy, Astrophysics and Space Science

Sapienza University of Rome

School of Aerospace Engineering

Automation, Robotics, and Controls for Aerospace Laboratory (ARCALab)

📅 November 2018 – December 2021

📍 Rome, IT

- Dissertation: Optimal Guidance and Control for Space Missions on Asteroids and Planetary Bodies
- Advisor: Prof. Fabio Curti
- Graduation grade: Excellent with honors

M.S. in Space & Astronautical Engineering

Sapienza University of Rome

📅 November 2015 – October 2017

📍 Rome, IT

- Dissertation: Study of Binary and Ternary Asteroids Systems Dynamics
- Advisor: Prof. Christian Circi
- Graduation grade: 110/110 with honors

B.S. in Aerospace Engineering

Sapienza University of Rome

📅 September 2012 – November 2015

📍 Rome, IT

- Dissertation: Study of New Horizons Gravity Assist Maneuver
- Advisor: Prof. Fabio Curti
- Graduation grade: 110/110 with honors

Musical Degree in Classical Piano

Licinio Refice Conservatory of Music

📅 October 2008 – July 2016

📍 Frosinone, IT

- Advisor: Prof. Roberto Cavalli
- Graduation grade: 8.5/10

EXPERIENCE

Researcher

Institute on AI for Space Sustainability

📅 March 2024 – Ongoing

📍 Tucson, AZ

- Developed a **light curve simulator for GEO objects** to build extensive datasets for neural networks training.
- Analyzed **physics-informed neural networks to characterize the space objects and their behavior**, with particular regard to the bus architecture, attitude and attitude rate estimation, from the visual magnitude data acquired by telescopes.

Postdoctoral Research Associate

The University of Arizona

Department of Systems & Industrial Engineering

Space Systems Engineering Laboratory (SSEL)

📅 February 2023 – Ongoing

📍 Tucson, AZ

- Developed and exploited **Physics-Informed Neural Networks frameworks to solve constrained optimal control problems** for aerospace applications, for example including missile autopilot, hypersonic trajectories, rendezvous with obstacle avoidance, attitude control.
- **Combined PINNs and Deep Theory of Functional Connections** to enhance the accuracy, optimality, and efficiency of Reinforcement Learning on aerospace optimal control. Investigated stability certification analysis of neural networks.
- Conceptualized and created a **novel framework to perform angle-only observations orbit determination of space objects using physics-informed neural networks** for space situational awareness purposes. The feasibility and accuracy of the method have been assessed through real observation data acquired by the Space4 Center telescopes. Test cases involved objects in the GEO belt and the X-GEO domain, including the cislunar environment.

Postdoctoral Research Associate

Massachusetts Institute of Technology

Department of Aeronautics and Astronautics

Astrodynamics, space Robotics, and Controls Laboratory (ARCLab)

📅 February 2022 – January 2023

📍 Cambridge, MA

- Conducted research on the **evolution of the space environment through source-sink evolutionary models and Monte Carlo models**. Several aspects were considered in the dynamical model design, such as the launch rate, collision avoidance, collision and explosion modeling, space weather effects, and the sinking mechanism of the atmospheric drag.
- Proposed a theoretical approach, based on the equilibrium points of a dynamical system, to **estimate the orbital capacity of the lower LEO environment** in terms of the maximum number of satellites that can reside in orbit while maintaining the space environment sustainable.

Visiting Researcher

The University of Arizona

Department of Systems & Industrial Engineering

Space Systems Engineering Laboratory (SSEL)

📅 January 2021 – July 2021

📍 Tucson, AZ

- Application of **Pontryagin Neural Networks to accurately learn the solution of optimal control problems with discontinuous control** exploiting the combination between PoNN, smoothing techniques, like the hyperbolic tangent method, and continuation procedures.
- Developed **Bellman Neural Networks (BeNNs) to obtain the solution of the Hamilton-Jacobi-Bellman partial differential equation** to retrieve the closed-loop optimal control.

Visiting Researcher

The University of Arizona

Department of Systems & Industrial Engineering

Space Systems Engineering Laboratory (SSEL)

📅 January 2020 – July 2020

📍 Tucson, AZ

- Designed a novel **physics-informed neural network framework, called Pontryagin Neural Networks (PoNNs), to tackle optimal control problems via indirect method** by leveraging my expertise regarding the Theory of Functional Connections to solve systems of ordinary differential equations. Several aerospace applications have been analyzed, with a particular focus on optimal orbit transfers and optimal landing trajectories on small and large planetary bodies.
- Designed **Reinforcement Meta-learning algorithms to compute safe optimal control for landing trajectories on the Moon**. The framework benefited from the interface with Blender, which is a ray tracing open source software that allows to realistically simulate the lunar terrain and the spacecraft camera. A sequence of successive images was fed to the neural network that provided the thrust control vector as output.
- Implemented **hazard detection and avoidance through neural networks** trained to identify and locate the safest regions to land on, far away from slopes and craters.

Research Scholarship

Sapienza University of Rome

Department of Astronautical, Electrical and Energetic Engineering

📅 April 2018 – July 2018

📍 Rome, IT

- Developed a software to **optimize the control maneuvers of spacecraft formation flying for station-keeping and reconfiguration purposes**. Metaheuristics algorithms have been exploited to retrieve the optimal control.

2018 ESA Summer School Scholarship

European Space Agency

📅 July 2018

📍 Alpbach, AT

- Improved my knowledge of space systems through lectures delivered by scientists and engineers from the European Space Agency. Together with my team, composed of a variety of Master's and Ph.D. students, I applied the acquired knowledge to perform a **preliminary mission design of a sample return mission from small solar system bodies**.
- **Analyzed the interplanetary trajectory optimization, the landing and ascend trajectories, as well as the attitude control of the spacecraft.** Thanks to this experience, I developed strong team building and team working skills, necessary to collectively accomplish the goals of a project. The results of this study have been presented at an international conference in planetary science [80] and have been published in a journal paper [12].

RESEARCH INTERESTS

- Optimal guidance and control for aerospace applications
- Astrodynamics
- Space situational awareness, orbit determination and space object behavioral analysis through light curves
- Space Robotics
- Dynamics around small planetary bodies
- Machine Learning, with particular focus on Physics-Informed Neural Networks.
- Functional interpolation applied to optimization (Theory of Functional Connections)

TEACHING

Teaching philosophy: "You can teach a student a lesson for a day, but if you can teach him to learn by creating curiosity, he will continue the learning process as long as he lives." (Clay P. Bedford).

Students' learning goal: Transfer the theoretical knowledge into practical skills by analyzing and solving real-world problems, drawn from recent academic papers and adapted to the students' knowledge level.

Teaching method: Retrieval practice at the beginning of the lesson, peer instruction to encourage discussion among students and to improve critical thinking and problem-solving skills.

Learning assessment: Quizzes, individual assignments and group project to develop both independent and teamwork skills.

Tutoring availability: Availability after class, during scheduled office hours, and by appointment for further clarifications about the concepts discussed in the classroom.

Teaching Assistant

Sapienza University of Rome
School of Aerospace Engineering

📅 February 2019 – May 2020

📍 Rome, IT

- **Attitude dynamics, determination, and control.** In this course, I contributed on testing and grading the students' for their final oral exam.
- **Advanced topics in aerospace engineering.** I delivered lectures and seminars about advanced topics in space engineering, mostly related to my personal research, including for example the study of lagrangian points and hovering trajectories around asteroids.
- **Space exploration robotic systems.** I delivered lectures on orbital dynamics, especially related to orbital perturbations, relative motion, circular restricted three-body problem, and the use of solar sails for proximity operations around asteroids.

MENTORING

- Graduate Students
 1. Agostinelli, I., *Artificial Intelligence applied to space surveillance and space traffic management*, Ph.D. Dissertation in Astronomy, Astrophysics, and Space Science, Sapienza University of Rome and Tor Vergata University of Rome, 2025. (**Co-advisor** of Ph.D.'s dissertation)
 2. Goracci, G., *Artificial intelligence techniques for the GNC system design of autonomous spacecraft for interplanetary missions*, Ph.D. Dissertation in Astronomy, Astrophysics, and Space Science, Sapienza University of Rome and Tor Vergata University of Rome, 2024. (**Co-advisor** of Ph.D.'s dissertation)
 3. Gupta, K., *Orbit Determination via DeepONet and RandONet*, University of Arizona, 2024. (Project **mentor**)
 4. Mastrofini, M., *Artificial intelligence techniques applied to onboard space navigation, surveillance and tracking*, Ph.D. Dissertation in Astronomy, Astrophysics, and Space Science, Sapienza University of Rome and Tor Vergata University of Rome, 2023. (**Co-advisor** of Ph.D.'s dissertation)
 5. Confido, A., *Cerebral Hemodynamics Integrated Estimation with PHysics (CHIEPH): A Physics-Informed Neural Network Approach for On-Board Health Monitoring of Astronauts*, University of Arizona, 2024. (Project **mentor**)
 6. Ramponi, L., *Optical Track-To-Track Correlation with Differential Algebra and Pontryagin Neural Networks*, Master's degree in Space Engineering, Politecnico of Milan, 2024. (**Co-advisor** of Master's thesis)
 7. Parker, W., *Effects of Thermosphere Contraction on Orbital Capacity in Low Earth Orbit*, Massachusetts Institute of Technology, 2022. (Project **mentor**)
 8. Pasiecznik, C., *Analysis of the Effects of the Launch Rate Distribution on the Stability of Source-Sink Orbital Debris Models*, Massachusetts Institute of Technology, 2022. (Project **mentor**)

9. Gusmini, D., *A new source-sink space environment evolution model including the effects of orbit transfer maneuvers*, Master's degree in Space Engineering, Politecnico of Milan, 2023. (**Co-advisor** of Master's thesis)
- Undergraduate Students
 1. Lersch, R., *Spectral Mixing for Resident Space Objects via the Solution of Mixed-Integer Nonlinear Programming Problems*, University of Arizona, 2024. (Project **mentor**)
 2. Pozzi, C., *Theory of Functional Connections applied to optimal maneuvers for autonomous in-orbit assembly*, Bachelor's degree in Aerospace Engineering, Sapienza University of Rome, 2020. (**Co-advisor** of Bachelor's thesis)
 3. Sena, F., *Study on interplanetary trajectories towards Uranus and Neptune*, Bachelor's degree in Aerospace Engineering, Sapienza University of Rome, 2020. (**Co-advisor** of Bachelor's thesis)
 4. Pacchiarini, M., *Soft Landing On Asteroids Via ZEM/ZEV Guidance*, Bachelor's degree in Aerospace Engineering, Sapienza University of Rome, 2021. (**Co-advisor** of Bachelor's thesis)
 5. Pagnozzi, P., *Neural networks application for asteroids gravity field estimation: the case of Gaspra 951*, Bachelor's degree in Aerospace Engineering, Sapienza University of Rome, 2020. (**Co-advisor** of Bachelor's thesis)
 6. Chelini, P. O., *Minimum-fuel low-thrust optimization*, Bachelor's degree in Aerospace Engineering, Sapienza University of Rome, 2021. (**Co-advisor** of Bachelor's thesis)

CERTIFICATIONS

- CIRTl Associate and Practitioner level certificates for professional development in teaching (The University of Arizona, 2024-2025).
- Qualification exam for the profession of Industrial Engineer. Section: Aerospace Engineering (Sapienza University of Rome, Italy, 2020).
- Satellite Tool Kit (STK): Basic Certification, Master Certification, Grand Master Certification.
- CESL English Oral Proficiency Interview: C2 Level (The University of Arizona, 2020).

ASSOCIATIONS

- AIAA: Member of the American institute of Aeronautics and Astronautics
- SIMCA: Extraordinary member of the Italian Society of Celestial Mechanics and Astrodynamics
- ISSNAF: Italian Scientists & Scholars in North America Foundation

AWARDS

- **Best Paper Award:** Best paper award at the conference 2nd International Conference on Applied Intelligence and Informatics 2022: Schiassi E., D'Ambrosio A., Furfaro R., 2022. " *Overview of X-TFC Applications for Aerospace Optimal Control Problems*" [50].
- **Scholarship for research abroad, 2021:** I was awarded a scholarship, assigned by Sapienza University of Rome, to spend a period abroad of six months at "The University of Arizona" as a visiting PhD student for the research project entitled "Study and Application of Machine Learning Algorithms for Real-time Optimal Control Problems for Autonomous Space Exploration Missions"
- **Excellent Graduate, 2018:** I was recognized as an Excellent Graduate by the Rector of Sapienza University of Rome, ranking among the top graduates in the Master's program across the entire university for the academic year 2016/2017.

PROJECTS & PROPOSALS

- **AI for Space Operations, Safety and Sustainability:** I contributed to write this international research proposal, submitted to and accepted by the UK Space Agency, about the acceleration of the development of AI technologies from conception, at early technology readiness levels, to adoption in the space sector.
- **Analysis of LEO Orbital Capacity via Source-Sink Evolutionary Models for National Security Applications and Decision Support:** I contributed to write this research proposal, submitted to and accepted by DARPA (tactical technology section), about the analysis of the Low Earth Orbit capacity and its consequences on U.S. national security applications and decision support.
- **Adaptive Space Governance and Decision-Support using Source-Sink Evolutionary Environmental Models:** I contributed to write this research proposal, submitted to and accepted by NASA, about the use of source-sink evolutionary environmental models to study the evolution of the space environment for space governance, policy design and decision-support.
- **SSE (Sun cubE onE: A Multi-wavelength Synoptic Solar Micro Satellite):** I worked on the mission analysis for this proposal for future cubesat missions which was submitted to and accepted by the Italian Space Agency, in collaboration with School of Aerospace Engineering of Sapienza University of Rome, Tor Vergata University of Rome, Italian National Institute of Nuclear Physics (INFN), Argotec, Next, Optec. The study behind this mission proposal has been published in a conference [75] and a journal paper [15].
- **SPOT (Star sensor image on-board Processing for Orbiting objects deTection):** I worked on this project in collaboration with School of Aerospace Engineering of Sapienza University of Rome and Italian Space Agency. Specifically, I worked on the ground software and contributed to write the systems requirements and the documentation associated to the deliverables.

REVIEWER FOR SCIENTIFIC JOURNALS AND CONFERENCES

- **Journals:** Advances in Space Research; Acta Astronautica; The Journal of The Astronautical Sciences; Journal of Guidance, Control, and Dynamics; Journal of Spacecraft and Rockets; Journal of Aerospace Information Systems; IEEE Transactions on Aerospace and Electronic Systems; Aerospace; Scientific Reports - Nature; Communications in Nonlinear Science and Numerical Simulation; Neural Computing and Applications; Wind Energy Science.

- **Conferences:** International Conference on Applied Intelligence and Informatics 2022; 2024 IEEE Congress on Evolutionary Computation (CEC) - Special Session: Computational Intelligence in Space and Aerospace.

PUBLICATIONS

Journal Papers

- [1] Ramponi L., **D'Ambrosio A.**, Cipollone R., De Riz A., Furfaro R., Reddy V., Di Lizia P., 2025. A Pontryagin Neural Network Application to Tracklets Correlation of Optical Observations, *Acta Astronautica*, 238 (2026), pp. 580-597.
<https://doi.org/10.1016/j.actaastro.2025.09.032>
- [2] Scorsoglio, A., **D'Ambrosio, A.**, Le Corre, L., Gray, B., Reddy, V., Furfaro, R., 2025. Physics-Informed Orbit Determination for X-GEO Space Situational Awareness, *Acta Astronautica*, 238 (2026), pp. 271-285.
<https://doi.org/10.1016/j.actaastro.2025.09.002>
- [3] **D'Ambrosio A.**, Benedikter B., and Furfaro R., 2025. Physics-Informed Pontryagin Neural Networks for Path-Constrained Optimal Control Problems, *Journal of Guidance, Control, and Dynamics*, Vol. 48, No. 8 (2025), pp. 1861-1877.
<https://doi.org/10.2514/1.G008854>
- [4] Jang, D., Gusmini, D., Siew, P.M., **D'Ambrosio, A.**, Servadio, S., Machuca, P., Linares, R., 2025. A New Monte-Carlo Model for the Space Environment. *Journal of Spacecraft and Rockets*.
<https://doi.org/10.2514/1.A36137>
- [5] Federici, L., **D'Ambrosio, A.**, Furfaro, R., and Reddy, V., 2024. Optimal Sensor Tasking for Space Domain Awareness Via a Beam A*-Search Algorithm. *The Journal of the Astronautical Sciences* 71, 34 (2024).
<https://doi.org/10.1007/s40295-024-00455-6>
- [6] Drozd K., Roberto F., and **D'Ambrosio A.**, 2024. A Theory of Functional Connections-Based hp-Adaptive Mesh Refinement Algorithm for Solving Hypersensitive Two-Point Boundary-Value Problems, *Mathematics*, 12(9):1360:1–1360:35.
<https://doi.org/10.3390/math12091360>
- [7] **D'Ambrosio A.**, and Linares R., 2024. Carrying Capacity of Low Earth Orbit Computed Using Source-Sink Models, *Journal of Spacecraft and Rockets*, Vol. 61, No. 6 (2024), pp. 1447-1463.
<https://doi.org/10.2514/1.A35729>
- [8] **D'Ambrosio A.**, and Furfaro R., 2024. Learning Fuel-Optimal Trajectories for Space Applications via Pontryagin Neural Networks, *Aerospace*, 11(3):228.
<https://doi.org/10.3390/aerospace11030228>
- [9] Gusmini D., **D'Ambrosio A.**, Servadio S., Siew P.M., Di Lizia P., Linares R., 2024. Effects of Orbit Raising and Deorbiting in Source-Sink Evolutionary Models, *Journal of Spacecraft and Rockets*.
<https://doi.org/10.2514/1.A35849>
- [10] Servadio S., Simha N., Gusmini D., Jang D., Francis T. St., **D'Ambrosio A.**, Lavezzi G., Linares R., 2023. A Risk Index For The Optimal Ranking Of Active Debris Removal Targets, *Journal of Spacecraft and Rockets*.
<https://doi.org/10.2514/1.A35752>
- [11] Drozd K., Furfaro R., Schiassi E., **D'Ambrosio A.**, 2023. Physics-Informed Neural Networks and Functional Interpolation for Solving the Matrix Differential Riccati Equation, *Mathematics*, 2023, 11(17), 3635.
<https://doi.org/10.3390/math11173635>
- [12] Ribeiro T., **D'Ambrosio A.**, Calabuig G.J.D., Athanasopoulos D., Bates H., Riegler C., Gassot O., Gerig S.B., Gómez-González J.L., Huber N., Seton R., Magalhães T.E.C., 2023. CARINA: A near-Earth D-type asteroid sample return mission, *Acta Astronautica*,
<https://doi.org/10.1016/j.actaastro.2023.07.035>
- [13] **D'Ambrosio A.**, Carbone A., Curti F., 2023. Optimal Maneuvers Around Binary Asteroids Using Particle Swarm Optimization and Machine Learning, *Journal of Spacecraft and Rockets*,
<https://doi.org/10.2514/1.A35317>
- [14] **D'Ambrosio A.**, Servadio S., Siew P.M., Linares R., 2023. A novel source-sink model for space environment evolution with orbit capacity assessment, *Journal of Spacecraft and Rockets*,
<https://doi.org/10.2514/1.A35579>
- [15] Giovannelli L., Berrilli F., Casolino M., Curti F., Del Moro D., Calchetti D., Cantoresi M., **D'Ambrosio A.**, Francisco G., Giobbi P., Marcelli L., Mazzotta P., Mugatwala R., Pucacco G., Reda R., Dhara S.K., Tombesi F., Blandino D., Benigno N., Cilia M., Di Salvo A., Di Tana V., Ingiosi F., Loddò S., Marmonti M., Musazzi M., Simonetti S., Truscelli G., 2022. Sun CubE OnE: A Multi-wavelength Synoptic Solar Micro Satellite, *Advances in Space Research*,
<https://doi.org/10.1016/j.asr.2022.09.044>
- [16] Schiassi E., **D'Ambrosio A.**, Furfaro R., 2022. Bellman Neural Networks for the Class of Optimal Control Problems with Integral Quadratic Cost, *IEEE Transactions on Artificial Intelligence*,
<https://doi.org/10.1109/TAI.2022.3206735>
- [17] **D'Ambrosio A.**, Henderson T., Clocchiatti A., Mortari D., 2022. Constrained Optimization of n-Impulse Periodic Close Encounter Orbits for Inspection Missions, *Advances in Space Research*, Volume 70, Issue 11, 1 December 2022, Pages 3393-3404
<https://doi.org/10.1016/j.asr.2022.08.049>
- [18] Federici L., Scorsoglio A., Ghilardi L., **D'Ambrosio A.**, Benedikter B., Zavoli A., Furfaro R., 2022. Image-based Meta-Reinforcement Learning for Autonomous Guidance of an Asteroid Impactor, *Journal of Guidance, Control and Dynamics*,
<https://doi.org/10.2514/1.G006832>
- [19] **D'Ambrosio A.**, Schiassi E., Johnston H., Curti F., Mortari D., Furfaro R., 2022. Time-Energy Optimal Landing on Planetary Bodies via Theory of Functional Connections, *Advances in Space Research*, Volume 69, Issue 12, June 2022, Pages 4198- 4220,
<https://doi.org/10.1016/j.asr.2022.04.009>

- [20] Schiassi E., **D'Ambrosio A.**, Drozd K. Curti F., Furfaro R., 2022. Physics-Informed Neural Networks for Optimal Planar Orbit Transfers, 2022, 59(3), *Journal of Spacecraft and Rockets*
<https://doi.org/10.2514/1.A35138>
- [21] Scorsoglio A., **D'Ambrosio A.**, Ghilardi L., Gaudet B., Curti F., Furfaro R., 2022. Image-based Deep Reinforcement Meta-Learning for Autonomous Lunar Landing, *Journal of Spacecraft and Rockets*, 59(1).
<https://doi.org/10.2514/1.A35072>
- [22] De Florio, M., Schiassi, E., **D'Ambrosio, A.**, Mortari, D. and Furfaro, R., 2021. Theory of Functional Connections Applied to Linear ODEs Subject to Integral Constraints and Linear Ordinary Integro-Differential Equations. *Mathematical and Computational Applications*, 26(3), p.65.
<https://doi.org/10.3390/mca26030065>
- [23] Schiassi, E., De Florio, M., **D'Ambrosio, A.**, Mortari, D. and Furfaro, R., 2021. Physics-informed neural networks and functional interpolation for data-driven parameters discovery of epidemiological compartmental models. *Mathematics*, 9(17), p.2069.
<https://doi.org/10.3390/math9172069>
- [24] **D'Ambrosio A.**, Carbone A., Spiller D., Curti F., 2021. PSO-Based Soft Lunar Landing with Hazard Avoidance: Analysis and Experimentation, *Aerospace*, 8(195).
<https://doi.org/10.3390/aerospace8070195>
- [25] **D'Ambrosio A.**, Spiller D., Curti F., 2021. Time suboptimal formation flying manoeuvres through improved magnetic charged system search, *Advances in Space Research*, 67(11), pp: 3462-3477.
<https://doi.org/10.1016/j.asr.2020.05.013>
- [26] **D'Ambrosio A.**, Schiassi E., Curti F., Furfaro R., 2021. Pontryagin Neural Networks with Functional Interpolation for Optimal Intercept Problems, *Mathematics*, 9(9), 996.
<https://doi.org/10.3390/math9090996>
- [27] **D'Ambrosio A.**, Spiller D., Curti F., 2020. Improved magnetic charged system search optimization algorithm with application to satellite formation flying *Engineering Applications of Artificial Intelligence*, 89.
<https://doi.org/10.1016/j.engappai.2020.103473>
- [28] Circi C., **D'Ambrosio A.**, Lei H., Ortore E., 2019. Global mapping of asteroids by frozen orbits: The case of 216 kleopatra, *Acta Astronautica*, 161, pp: 101- 107.
<https://doi.org/10.1016/j.actaastro.2019.05.026>
- [29] **D'Ambrosio A.**, Circi C., Zeng X., 2019. Solar Photon Sail Hovering Orbits about Single and Binary Asteroids, *Advances in Space Research*, 63(11), pp: 3691- 3705.
<https://doi.org/10.1016/j.asr.2019.02.021>

Conference Proceedings

- [27] Scorsoglio A., **D'Ambrosio A.**, Furfaro R., 2025. Stability Certification of Reinforcement Learning-Based Control for Aerospace Applications, 2025 AAS/AIAA Space Flight Mechanics Meeting, 19-23 January 2025, Kaua'i, Hawaii, USA
- [28] Benedikter B., **D'Ambrosio A.**, Scorsoglio A., Furfaro R., Vasile M., 2025. Physics-Informed Machine Learning for Attitude Estimation from Light Curves, 2025 AAS/AIAA Space Flight Mechanics Meeting, 19-23 January 2025, Kaua'i, Hawaii, USA
- [29] **D'Ambrosio A.**, Scorsoglio A., Furfaro R., Reddy V., 2025. Physics-Informed Orbit Determination for X-GEO Objects Maneuver Estimation: the Peregrine Spacecraft Case Study, 2025 AAS/AIAA Space Flight Mechanics Meeting, 19-23 January 2025, Kaua'i, Hawaii, USA
- [30] **D'Ambrosio A.**, Furfaro R., 2025. Pontryagin Neural Networks and Artificial Potential Functions for Safe Trajectory Optimization, 2025 AAS/AIAA Space Flight Mechanics Meeting, 19-23 January 2025, Kaua'i, Hawaii, USA
- [31] Benedikter A., **D'Ambrosio A.**, Furfaro R., 2025. Rocket Ascent Trajectory Optimization via Physics-Informed Pontryagin Neural Networks, 2025 AIAA Science and Technology Forum and Exposition (AIAA SciTech Forum), 6 January - 10 January 2025, Hyatt Regency Orlando, 9801 International Drive, Orlando, Florida, USA
- [32] **D'Ambrosio A.**, Benedikter A., Furfaro R., 2025. Physics-Informed Pontryagin Neural Networks for Path-Constrained Optimal Control Problems, 2025 AIAA Science and Technology Forum and Exposition (AIAA SciTech Forum), 6 January - 10 January 2025, Hyatt Regency Orlando, 9801 International Drive, Orlando, Florida, USA
- [33] Ramponi L., **D'Ambrosio A.**, Cipollone R., De Riz A., Furfaro R., Reddy V., Di Lizia P., 2024. A Pontryagin Neural Network Application to Tracklets Correlation of Optical Observations, 75th International Astronautical Congress (IAC), Milan, Italy, October 18-24, 2024
- [34] **D'Ambrosio A.**, Scorsoglio A., Battle A., Furfaro R., Reddy V., 2024. Machine Learning-based Light Curves Brightness Prediction for Space Objects in the Geostationary Belt, 2024 AIAA Science and Technology Forum and Exposition (AIAA SciTech Forum), 8 January - 12 January 2024, Hyatt Regency Orlando, 9801 International Drive, Orlando, Florida, USA
- [35] Scorsoglio A., **D'Ambrosio A.**, Cambell T., Furfaro R., Reddy V., 2024. Orbit determination pipeline for geostationary objects using physics-informed neural networks, 2024 AIAA Science and Technology Forum and Exposition (AIAA SciTech Forum), 8 January - 12 January 2024, Hyatt Regency Orlando, 9801 International Drive, Orlando, Florida, USA
- [36] Furfaro R., **D'Ambrosio A.**, 2024. Increasing Autonomy of Aerospace Systems via PINN-based Solutions of HJB Equation, 2024 AIAA Science and Technology Forum and Exposition (AIAA SciTech Forum), 8 January - 12 January 2024, Hyatt Regency Orlando, 9801 International Drive, Orlando, Florida, USA
- [37] Federici L., **D'Ambrosio A.**, Furfaro R., Reddy V., 2023. Optimal Sensor Tasking for Space Domain Awareness via a Beam A*-Search Algorithm, *Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference*, Maui, Hawaii (USA), September 19-22
- [38] Scorsoglio A., **D'Ambrosio A.**, Ghilardi L., Furfaro R., Reddy V., 2023. Physics-Informed Orbit Determination for Cislunar Space Applications, *Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference*, Maui, Hawaii (USA), September 19-22
- [39] **D'Ambrosio A.**, Scorsoglio A., Ghilardi L., Furfaro R., Vishnu R., 2023. State estimation with angle-only observations in cislunar space

- via physics-informed neural network, *2023 AAS/AIAA Astrodynamics Specialist Conference, Big Sky, Montana, 13-17 August*
- [40] Federici L., **D'Ambrosio A.**, Furfaro R., Vishnu R., 2023. An A*-Search Approach to Optimal Telescope Tasking for Space Domain Awareness, *2023 AAS/AIAA Astrodynamics Specialist Conference, Big Sky, Montana, 13-17 August*
 - [41] Goracci G., **D'Ambrosio A.**, Curti F., 2023. Machine Learning-aided Satellite Navigation through Extended Kalman Filter using Magnetometer Measurements, *2023 AAS/AIAA Astrodynamics Specialist Conference, Big Sky, Montana, 13-17 August*
 - [42] Lifson M., **D'Ambrosio A.**, Jang D., Gusmini D., Pasiecznik C., Linares R., 2023. MOCAT-SSEM: A Source-Sink Evolutionary Model for Space Debris Environment Evolutionary Modeling, *9th Annual Space Traffic Management Conference 2023, Austin, Texas, March 01-02*
 - [43] Gusmini D., **D'Ambrosio A.**, Servadio S., Siew P.M., Di Lizia P., Linares R., 2023. The effects of orbit raising and decay in orbital capacity models, *33rd AAS/AIAA Space Flight Mechanics Meeting, Austin, Texas, January 15-19*
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 - [50] Schiassi E., **D'Ambrosio A.**, Furfaro R., 2022. Overview of X-TFC Applications for Aerospace Optimal Control Problems, *2nd International Conference on Applied Intelligence and Informatics 2022 – University Mediterranea of Reggio Calabria, Reggio Calabria, IT, 01-03 September (Best Paper Award)*.
 - [51] **D'Ambrosio A.**, Servadio S., Siew P.M., Jang D., Lifson M., Linares R., 2022. Analysis of the LEO orbital capacity via probabilistic evolutionary model, *2022 AAS/AIAA Astrodynamics Specialist Conference – Charlotte (NC), USA, 07-11 August*
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 - [53] Furfaro R., **D'Ambrosio A.**, Schiassi E., Drozd K., Scorsoglio A., 2022. Physics-Informed Neural Networks for Closed-Loop Guidance and Control in Aerospace Systems, *In AIAA SciTech Forum 2022, San Diego (CA), USA, 3-7 January*
 - [54] Federici L., Scorsoglio A., Ghilardi L., **D'Ambrosio A.**, Benedikter B., Zavoli A., Furfaro R., 2022. Reinforcement Meta-Learning for Autonomous Image-Based Guidance around Binary Asteroid Systems, *In AIAA SciTech Forum 2022, San Diego (CA), USA, 3-7 January 2022*
 - [55] Carbone A., Agostinelli I., **D'Ambrosio A.**, Curti F., 2021. Optimization of Hopping Trajectories for Asteroids Surface Exploration, *72nd International Astronautical Congress 2021, Dubai, United Arab Emirates 25-29 October*
 - [56] **D'Ambrosio A.**, Schiassi E., Curti F., Furfaro R., 2021. Physics-Informed Neural Networks for Optimal Proximity Maneuvers with Collision Avoidance around Asteroids, *2021 AAS/AIAA Astrodynamics Specialist Conference – Big Sky, Montana, 09-11 August*
 - [57] **D'Ambrosio A.**, Conforti G., Agostinelli I., Curti F., 2021. Adaptive control for proximity maneuvers around asteroids using Extreme Learning Machine-based gravity field estimation, *31st AAS/AIAA Space Flight Mechanics Meeting, Virtual, Charlotte, North Carolina (USA)*
 - [58] **D'Ambrosio A.**, Carbone A., Mastrofini M., Curti F., 2021. Optimal reference orbit tracking around asteroids via Particle Swarm Optimization and inverse dynamics technique, *31st AAS/AIAA Space Flight Mechanics Meeting, Virtual, Charlotte, North Carolina (USA)*
 - [59] Sena F., **D'Ambrosio A.**, Curti F., 2021. Study on Interplanetary Trajectories towards Uranus and Neptune, *31st AAS/AIAA Space Flight Mechanics Meeting, Virtual, Charlotte, North Carolina (USA)*
 - [60] **D'Ambrosio A.**, Schiassi E., Curti F., Furfaro R., 2021. Physics-Informed Neural Networks Applied to a Series of Constrained Space Guidance Problems, *31st AAS/AIAA Space Flight Mechanics Meeting, Virtual, Charlotte, North Carolina (USA)*
 - [61] Schiassi E., **D'Ambrosio A.**, Scorsoglio A., Furfaro R., Curti F., 2021. Class of Optimal Space Guidance Problems solved via Indirect Methods and Physics-Informed Neural Networks, *31st AAS/AIAA Space Flight Mechanics Meeting, Virtual, Charlotte, North Carolina (USA)*
 - [62] Ghilardi L., **D'Ambrosio A.**, Scorsoglio A., Furfaro R., Curti F., 2021. Image-Based Lunar Landing Hazard Detection via Deep Learning, *31st AAS/AIAA Space Flight Mechanics Meeting, Virtual, Charlotte, North Carolina (USA)*
 - [63] Schiassi E., **D'Ambrosio A.**, Furfaro R., Curti F., 2020. Physics-Informed Neural Networks for optimal Intercept Problem, *3-rd IAA-AAS SciTech Forum 2020 on Space Flight Mechanics and Space Structures, Moscow, Russia*
 - [64] Scorsoglio A., **D'Ambrosio A.**, Ghilardi L., Furfaro R., Gaudet B., Linares R., Curti F., 2020. Safe lunar landing via images: a reinforcement meta-learning implementation for autonomous hazard avoidance and landing, *In 2020 AAS/AIAA Astrodynamics Specialist Conference, Lake Tahoe, CA, USA*
 - [65] Ghilardi L., **D'Ambrosio A.**, Scorsoglio A., Furfaro R., Linares R., Curti F., 2020. Image-based optimal powered descent guidance via deep recurrent imitation learning, *In 2020 AAS/AIAA Astrodynamics Specialist Conference, Lake Tahoe, CA, USA*
 - [66] Schiassi E., **D'Ambrosio A.**, Johnston H., Furfaro R., Curti F., Mortari D., 2020. Complete energy optimal landing on small and large planetary bodies via theory of functional connections, *In 2020 AAS/AIAA Astrodynamics Specialist Conference, Lake Tahoe, CA, USA*
 - [67] Schiassi E., **D'Ambrosio A.**, Johnston H., De Florio, M., Drozd, K., Furfaro, R., Curti, F. and Mortari, D., 2020. Physics-informed extreme theory of functional connections applied to optimal orbit transfer. *In Proceedings of the AAS/AIAA Astrodynamics Specialist Conference, Lake Tahoe, CA, USA (pp. 9-13)*.
 - [68] **D'Ambrosio A.**, Spiller D., Curti F., 2019. A self-adaptive magnetic charged system search for time- suboptimal formation flying manoeuvres, *International Workshop on Satellite Constellations and Formation Flying (IWSCFF), University of Strathclyde, Glasgow, July*

Posters & Presentations

- [69] Lersch, R., Campbell, T., **D'Ambrosio A.**, Battle, A., Pearson, N.C., Furfaro, R., Reddy, V. 2025. Spectrally Dominated Satellite Buses as a Test-Case for Spectral Unmixing Algorithms, *AFRL 6th Annual Voltron Meeting – 2025, 4-6 February 2025, Albuquerque, New Mexico (USA)*
- [70] **D'Ambrosio A.**, Benedikter B., Reddy V., Furfaro R., 2025. H-G System Light Curve Representation for Artificial GEO Objects, *AFRL 6th Annual Voltron Meeting – 2025, 4-6 February 2025, Albuquerque, New Mexico (USA)*
- [71] **D'Ambrosio A.**, Scorsoglio A., Furfaro R., 2024. Leveraging Physics-Informed Neural Networks for Precise Orbit Determination: A New Frontier in Space Situational Awareness, *Los Alamos - Arizona Days, Session: Space Exploration (Solar System Studies), 7-8 November 2024, University of Arizona*
- [72] Benedikter, B., **D'Ambrosio, A.**, Furfaro, R., 2024. Physics-Informed Neural Networks for Light Curve Analysis, *2nd International Workshop on AI for Space Sustainability, October 30th – November 1st, London (UK)*
- [73] **D'Ambrosio, A.**, Benedikter, B., Furfaro, R., 2024. The VerSSA Platform for Decentralized Privacy Preserving AI, *2nd International Workshop on AI for Space Sustainability, October 30th – November 1st, London (UK)*
- [74] Benedikter, B., **D'Ambrosio, A.**, Furfaro, R., 2024. Physics-Informed Neural Networks for Light Curve Analysis, *1st International Workshop on AI for Space Sustainability, June 24th – 25th, Glasgow (UK)*
- [75] Berrilli F., Giovannelli L., Casolino M., Curti F., Del Moro D., Calchetti D., **D'Ambrosio A.**, Marcelli L., Mazzotta P., Pucacco G., Tombesi F., Blandino D., Benigno N., Cilia M., Di Salvo A., Di Tana V., Ingiosi F., Loddo S., Marmonti M., Musazzi M., Simonetti S., Truscelli G. The Sun cubE onE (SEE) CubeSat mission, *Workshop "L'Impegno Italiano nel Settore dei CubeSat: Tecnologie e Missioni Future", Italian Space Agency, 05-07 July 2022, Rome, Italy*
- [76] **D'Ambrosio A.**. Pontryagin Neural Networks applied to optimal control problems for aerospace applications, *19th U.S. national congress on theoretical and applied mechanics, 19-24 June 2022, Austin, Texas*
- [77] Schiassi E., Leake C., De Florio M., Johnston H., **D'Ambrosio A.**, Furfaro R., Mortari D., 2020, May. Extreme Theory of Functional Connections: A Physics-Informed Neural Network Framework For Solving Forward and Inverse Problems Involving Parametric Differential Equations, *First World Online Conference on Theory of Functional Connections*
- [78] **D'Ambrosio A.**, Schiassi E., De Florio M., Curti F., Furfaro R., 2020. Physics-Informed Solutions of Optimal Control Problems via Extreme Theory of Functional Connections, *Los Alamos - Arizona Days, AI/ML/Data Science session, 18-19 May 2020, Online conference*
- [79] Curti F., **D'Ambrosio A.**, Agostinelli I., Mastrofini M., Facchinetti C., Ansalone L., Tuozzi A., 2020, January. Star Sensor Image On-Board Processing for Orbiting Objects Detection - SPOT, *5th IAA Conference on University Satellite Missions and CubeSat Workshop, Palazzo Rospigliosi, Rome*
- [80] Ribeiro T., Athanasopoulos D., Bates H., Bratli E., Breedveld M. J., **D'Ambrosio A.**, Calabuig G. J. D., Gassot O., Gerig S.-B., Gómez-González J. L., Haidari F., Huber N., Martin M., Riegler C. and Seton R., 2019, September. CARINA: A sample return mission concept to a near-Earth D-type asteroid, *European Planetary Science Congress (EPSC), Centre International de Conférences de Genève (CICG), Geneva*

Preprints

- [81] Pasiecznik C., **D'Ambrosio, A.**, Jang D., and Linares, R., 2022. A Dynamical Systems Analysis of the Effects of the Launch Rate Distribution on the Stability of a Source-Sink Orbital Debris Model. *arXiv preprint arXiv:2212.01000v1*. <https://doi.org/10.48550/arXiv.2212.01000>
- [82] **D'Ambrosio, A.**, Lifson, M. and Linares, R., 2022. The Capacity of Low Earth Orbit Computed using Source-sink Modeling. *arXiv preprint arXiv:2008.05554*. <https://doi.org/10.48550/arXiv.2206.05345>
- [83] Schiassi, E., **D'Ambrosio, A.**, De Florio, M., Furfaro, R. and Curti, F., 2020. Physics-informed extreme theory of functional connections applied to data-driven parameters discovery of epidemiological compartmental models. *arXiv preprint arXiv:2008.05554*. <https://doi.org/10.48550/arXiv.2008.05554>

OTHER COURSES

- **Saperi&Co Enterprise Incubation Course, 2018:** in collaboration with Sapienza University of Rome and the LVenture Group Investment Fund. The aim of the course Saperi&Co was to help recent graduates to develop their business idea and transform it into an enterprise project. During the course, I developed skills in many fields, such as team handling, company work management by using Scrum methodology, business, market analysis, competitive analysis, web-marketing
- **Astronomy Course, 2015:** During this course, I developed skills in astronomy and the basics of the use of telescopes.

SKILLS

- Matlab: Advanced
 - Python: Intermediate
 - Fortran: Beginner
 - C: Beginner
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- Windows: Advanced
- Microsoft Office: Advanced
- Linux (Ubuntu): Intermediate
- $\text{\LaTeX}$: Advanced

LANGUAGES

- English: Fluent
- Italian: Fluent